

HONOURS RESEARCH PROJECT

**Ideation Document:
Selection Pressure as an Alternative to Hand-Crafted Reward
Penalties**

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July 23, 2026



An ideation document submitted to the Faculty of Science, University of the Witwatersrand, Johannesburg, in partial fulfilment of the requirements for the degree of Bachelor of Science with Honours

Abstract

This ideation document outlines a research direction that asks whether *selection pressure* can substitute for reward engineering in safe reinforcement learning. Rather than tuning penalty weights or hand-crafting cost functions, the proposed approach swaps tuning for search: a JEPA-based world model imagines candidate plans, and a multi-objective evolutionary planner trades off task achievement against structural safety signals intrinsic to the model. The aim is to recover safety-performance trade-offs competitive with penalty-tuned and constrained RL methods, without committing to a scalarised objective before training begins.

Declaration

I, Sachin Mohan (Student Number: 2699183), hereby declare the contents of this ideation document to be my own work. This document is submitted for the degree of Bachelor of Science with Honours in Computer Science at the University of the Witwatersrand. This work has not been submitted to any other university, or for any other degree.

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1 Project Trajectory

The guiding question for this project is whether *selection pressure can substitute for reward engineering*. In other words: can the need to carefully tune safety penalties, or to hand-craft cost functions and constraint thresholds, be replaced by a search process that surfaces achievable safety-performance trade-offs automatically?

The intended trajectory is to swap *tuning for search*. A world model imagines candidate plans in its latent space; a multi-objective evolutionary planner then explores the trade-off between task achievement and safety, returning a Pareto front rather than a single scalarised solution. The operating point (how cautious the agent should be) becomes a deployment-time choice, not a hyperparameter fixed before learning begins.

2 What Has Already Been Done

World models have already been used for planning in latent space. In particular, hierarchical planning with latent world models has shown that multi-scale latent dynamics support effective model-predictive control without task-specific rewards or hierarchical policies [Zhang *et al.* 2026]. However, to the author’s knowledge, no recent work has combined online planning with evolutionary multi-objective methods in this setting: existing planners typically optimise a single scalar objective (for example via the cross-entropy method), and do not expose a Pareto front of safety-performance trade-offs derived from structural signals of the world model itself.

3 Research Topic

This research explores whether multi-objective evolutionary algorithms, operating in the latent space of a JEPA-based world model, can produce safe reinforcement learning behaviour using structural signals instead of hand-crafted cost functions.

4 Research Problem

Safe RL agents are predominantly trained by augmenting the task reward with a penalty term that discourages unsafe behaviour. This approach is fragile in practice: if the penalty weight is too small, the agent learns to ignore it and behaves unsafely; if it is too large, the agent becomes overly conservative and fails to complete the task at all. The correct weight depends on the environment dynamics, the task structure, and the relative scale of the reward and cost. These quantities are rarely known in advance, meaning the right trade-offs are found only through trial and error.

Constrained RL approaches (CPO, Lagrangian methods) are often proposed as an alternative, but they do not really solve this problem. Instead they relocate it: rather than tuning the penalty weight, the designer must now specify a cost function and a constraint threshold, both of which suffer from the same specification difficulty as the

reward itself. Writing a good cost function is frequently as hard as writing a good reward function.

The underlying issue is that both approaches require collapsing safety and task performance into a single scalar objective (or a fixed constraint) *before* training begins, forcing the designer to commit to a specific safety-performance trade-off in advance, without knowing what trade-offs are even achievable in a given environment.

5 Research Aim

The aim of this research is to investigate whether a multi-objective evolutionary planner, operating in the latent space of a JEPA-based world model and guided only by structural safety signals derived from the model itself, can recover safety-performance trade-offs competitive with hand-tuned penalty-based methods and constrained RL, without requiring manual specification of a cost function or penalty weight.

6 Method

The work proceeds in four phases.

1. A JEPA-based world model (LeWM) is reproduced on the Reacher and Push-T environments, providing a frozen latent-space substrate for planning.
2. The standard cross-entropy method planner is replaced with a multi-objective evolutionary planner based on NSGA-II, first verified on a single task-performance objective, then extended to jointly optimise task performance and safety, returning a Pareto front of candidate plans rather than a single scalarised solution.
3. A suite of structural safety objectives is developed from properties intrinsic to the world model itself: prediction surprise (as an out-of-distribution signal), latent energy and jerk (as smoothness signals), and small learned probes for safety-relevant physical quantities, rather than a hand-crafted cost function.
4. The resulting framework is evaluated against penalty-tuned CEM, constrained RL baselines (CPO, Lagrangian PPO), and ROSARL, using hypervolume and inverted generational distance to compare Pareto fronts, alongside a violation-of-expectation study using controlled perturbations to test whether prediction surprise reliably indicates safety violations.

References

[Zhang *et al.* 2026] Wancong Zhang, Basile Terver, Artem Zholus, Soham Chitnis, Harsh Sutaria, Mido Assran, Randall Balestriero, Amir Bar, Adrien Bardes, Yann LeCun, and Nicolas Ballas. Hierarchical planning with latent world models. *arXiv preprint arXiv:2604.03208*, 2026.